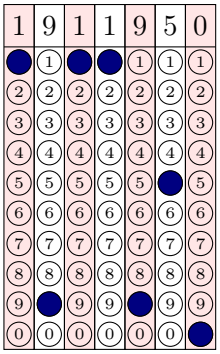
TEST VARIANTI RAQAMI: **1911950**

MATEMATIKA

Ushbu test varianti 30 ta test topshirig‘idan iborat. Uni bajarishga 1 soat vaqt belgilash tavsiya etiladi. Natijani bilish uchun javoblaringizni javoblar varaqasiga belgilab, rasmini **@dtmxizmatbot** orqali jo‘nating. Sizga omad tilaymiz!

TEST VARIANTI RAQAMINI JAVOBLAR VARAQASIGA TO‘G‘RI KO‘CHIRING!

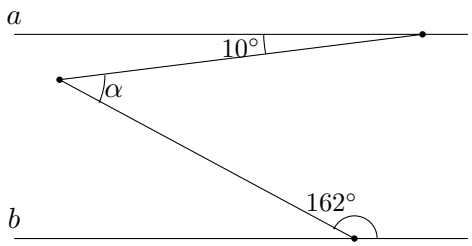


1. Agar $f(x) = \dfrac{x^3 - 4x^2 + x - 4}{x - 4}$ bo‘lsa, $f'(-2) + f(1)$ ning qiymatini toping.
A) -2 B) 2 C) -6 D) 4

2. (a_n) arifmetik progressiya uchun $a_{13} = 106$ va $S_{13} = 754$ bo‘lsa, a_3 ni toping.
A) *34* B) *28* C) *22* D) *26*

3. Asoslarining radiuslari 2 va 6 ga, o‘q kesimining diagonali 10 ga teng bo‘lgan kesik konus yon sirtining yuzini toping.
A) $15\sqrt{13}\pi$ B) $24\sqrt{13}\pi$ C) $16\sqrt{13}\pi$
D) $12\sqrt{13}\pi$

4. a va b parallel to‘g‘ri chiziqlar. Rasmdan foydalanib α burchakning qiymatini toping.



A) 28° B) *aniqlab bo‘lmaydi* C) 18°
D) 24°

5. $y = -3x + 7$ chiziqli funksiyaning y=1 to‘g‘ri chiziqqa nisbatan simmetrigini toping.
A) $y = 3x - 5$ B) $y = -3x - 7$
C) $y = -3x + 9$ D) $y = 3x + 1$

6. Tenglamani yeching:
$3 \cdot 2^{x-2} - 5 \cdot 2^{x-4} = 18 - 2^{x-3}$
A) *3* B) *5* C) *4* D) -2

7. $f(x) = (x+1)^2 + 4$ parabola uchining koordinatalarini toping.
A) $(-1; -4)$ B) $(-1; 4)$ C) $(1; -4)$
D) $(1; 4)$

8. Oltita sonning yig‘indisi 70 ga teng. Ulardan birini 2 marta oshirib, qolganlarini o‘zgartirmagan holda yig‘indisi 78 ga teng bo‘lsa, o‘zgartirilmagan beshta sonning yig‘indisini toping.
A) *66* B) *63* C) *67* D) *62*

9. Teng yonli uchburchakning asosi yon tomonidan 6 ga ortiq. Uchburchakning asosiga tushirilgan balandligi 6 ga teng bo‘lsa, uning asosini toping.
A) *10* B) *24* C) *16* D) *18*

10. $y = \ln\left(\dfrac{5x+6}{2x+15}\right)$ funksiyaning grafigiga abssissasi $x_0 = 3$ bo‘lgan nuqtadan urinma o‘tkazilgan. Bu urinma va koordinata o‘qlari hosil qilgan uchburchakning yuzini toping.
A) $\dfrac{3}{14}$ B) $\dfrac{3}{7}$ C) $\dfrac{9}{14}$ D) $\dfrac{9}{7}$

11. Hisoblang: $\sqrt{2019^2 - 2017 \cdot 2021}$
A) *2* B) *12* C) *8* D) *18*

12. Ikkidan katta bo‘lgan barcha tub sonlarni 4 ga bo‘lganda qanday qoldiqlar qoladi?
A) *1, 3* B) *0, 1, 2, 3* C) *2, 3* D) *1, 2, 3*

13. $(7; -12)$ nuqtaning ordinatalar o‘qiga nisbatan simmetrik bo‘lgan nuqtasini toping.
A) *(12; −7)* B) *(7; 12)* C) *(−7; −12)*
D) *(−7; 12)*

14. $(x+1) \cdot (|x| - 1) \geq 2$ tengsizlikni yeching.
A) $\left[\sqrt{3}; +\infty\right)$ B) $(-\infty; 0] \cup \left[\sqrt{3}; +\infty\right)$
C) $\left(-\infty; -\sqrt{3}\right] \cup \left[\sqrt{3}; +\infty\right)$
D) $\left(-\infty; -\sqrt{3}\right]$

15. Hisoblang: $\left(7\dfrac{17}{36} - 9\dfrac{7}{12}\right) : \dfrac{2}{9} - \dfrac{3}{26} \cdot 4\dfrac{1}{3} + 3$
A) -6 B) 1 C) *13* D) -7

© Davlat test markazi, 2019

16. Hisoblang: $\dfrac{\sqrt{32}+\sqrt{50}-\sqrt{72}}{\sqrt{18}}$

A) 2 B) 3 C) 1 D) $\dfrac{\sqrt{5}}{3}$

17. Hisoblang: $\dfrac{4,3253^2-5,6747^2}{5,6747-4,3253}$

A) *10* B) *9* C) *−10* D) *−1,3494*

18. Ifodani soddalashtiring ($x>0$):

$$\frac{2}{x\,(x+2)}+\frac{2}{(x+2)\,(x+4)}+\frac{2}{(x+4)\,(x+6)}$$

A) $\dfrac{6}{x^2+6x}$ B) $\dfrac{24}{x^2+6x}$ C) $\dfrac{12}{x^2+6x}$

D) $\dfrac{2x+12}{x^2+6x}$

19. Agar $f\,(x)=\log_2 x^3+1$ bo‘lsa,

$f\,(2)+f\left(\dfrac{1}{x}\right)=f\,(x)$ tenglamani yeching.

A) 2 B) $2\sqrt{2}$ C) $\sqrt[4]{8}$ D) $\sqrt[3]{4}$

20. $A=\{a;\,b;\,c;\,d;\,e;\,f\}$ to‘plamning elementlari soni 2 ta bo‘lgan nechta qism to‘plami bor?

A) *4* B) *15* C) *8* D) *16*

21. Hisoblang: $\dfrac{16\sin 251^\circ-12\cos 161^\circ}{2\cos 19^\circ}$.

A) 3 B) −2 C) 2 D) −3

22. 6 ta to‘g‘ri chiziqlar ko‘pi bilan nechta nuqtada kesishadi?

A) *15* B) *21* C) *12* D) *28*

23. Ifodani soddalashtiring:

$$-\frac{9a^4b^3}{16c^3d^2}\cdot\left(-2\frac{2}{3}\cdot\frac{c^3d}{a^3b^2}\right)$$

A) $-\dfrac{3a^2b}{2d}$ B) $\dfrac{3ab}{2d}$ C) $-\dfrac{3ab}{2d}$ D) $\dfrac{3a^2b}{2d}$

24. Tenglamani yeching:

$(\sin\,(x+1,5\pi)-1)\cdot \operatorname{tg}\dfrac{x}{2}=0.$

A) $\pi+2\pi k,\ k\in Z$ B) $\pi k,\ k\in Z$

C) $2\pi k,\ k\in Z$ D) $\dfrac{\pi}{2}+\pi k,\ k\in Z$

25. $\int \dfrac{1}{\sin^2(2x-3)}dx$ integralni hisoblang.

A) $\dfrac{ctg\,(3-2x)}{3}+C$ B) $-\dfrac{ctg\,(3-2x)}{2}+C$

C) $-\dfrac{ctg\,(3-2x)}{3}+C$ D) $\dfrac{ctg\,(3-2x)}{2}+C$

26. $2\sqrt[4]{x\cdot\left(7+4\sqrt{3}\right)}\cdot\sqrt{2\sqrt{x}-\sqrt{3x}}=x$ tenglama ildizlarining o‘rta arifmetik qiymatini toping.

A) *4* B) *2* C) *0* D) *1*

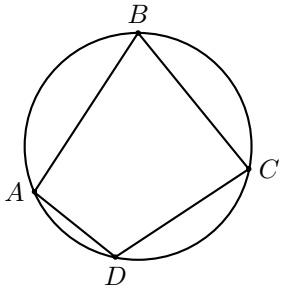
27. $\dfrac{\left(x^4-16\right)\sqrt{12-x-x^2}}{\sqrt{-x}}=0$ tenglamaning haqiqiy ildizlari ko‘paytmasini toping.

A) 6 B) 16 C) 8 D) −8

28. Agar to‘g‘ri silindr o‘q kesimining yuzi 8 ga teng bo‘lsa, silindrning yon sirti yuzini toping.

A) 4π B) 8π C) 2π D) 16π

29. Rasmda $ABCD$ to‘rtburchak va unga tashqi chizilgan aylana tasvirlangan. Agar AB=5, BC=4, CD=3 va AD=2 bo‘lsa, $\angle ABC$ ning kosinusini toping.



A) $\dfrac{8}{13}$ B) $\dfrac{6}{13}$ C) $\dfrac{5}{13}$ D) $\dfrac{7}{13}$

30. Mis, rux va qo‘rg‘oshindan iborat bo‘lgan 15 kg li qotishmaning tarkibida 20% mis bo‘lib, rux va qo‘rg‘oshinning og‘irliklari mos ravishda 5:1 nisbatda. Qotishmadagi ruxning og‘irligi misning og‘irligidan necha kilogrammga ko‘p?

A) *7* B) *4,5* C) *5* D) *6*

Test varianti “Test topshiriqlari to‘plami 2019” asosida shakllantirilgan:

1 – 80-bet 34	7 – 72-bet 33	13 – 116-bet 15	19 – 56-bet 7	25 – 85-bet 3
2 – 33-bet 36	8 – 16-bet 6	14 – 67-bet 12	20 – 119-bet 3	26 – 65-bet 62
3 – 114-bet 58	9 – 96-bet 47	15 – 15-bet 57	21 – 38-bet 35	27 – 59-bet 6
4 – 93-bet 22	10 – 83-bet 57	16 – 23-bet 11	22 – 123-bet 3	28 – 110-bet 27
5 – 74-bet 49	11 – 28-bet 60	17 – 10-bet 1	23 – 47-bet 23	29 – 104-bet 103
6 – 53-bet 12	12 – 9-bet 22	18 – 49-bet 42	24 – 44-bet 96	30 – 21-bet 44

© Davlat test markazi, 2019